

CRADOCK, South Africa

WMO: 687440

Lat: 32.1675S

Lon: 25.6253E

Elev: 926

StdP: 90.68

Time Zone: 2.00 (E02)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 0.4	(c) 1.8	(d) -11.7	(e) 1.5	(f) 14.3	(g) -9.9	(h) 1.8	(i) 15.2	(j) 9.8	(k) 20.9	(l) 8.6	(m) 18.6	(n) 2.7	(o) 340	(p) 0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 13.9	(c) 35.6	(d) 17.1	(e) 33.7	(f) 16.8	(g) 31.9	(h) 16.7	(i) 19.8	(j) 28.0	(k) 19.2	(l) 27.3	(m) 18.7	(n) 26.6	(o) 4.6	(p) 310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 17.7	(b) 14.2	(c) 22.0	(d) 17.0	(e) 13.6	(f) 21.7	(g) 16.4	(h) 13.1	(i) 21.4	(j) 61.0	(k) 27.8	(l) 58.9	(m) 27.3	(n) 57.2	(o) 26.7	(p) 22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.0	(b) 9.0	(c) 8.1	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 10.0	(b) 9.0	(c) 8.1	(d) WB	(e) -3.7	(f) 21.0	(g) 1.1	(h) 0.8	(i) -4.5	(j) 21.5	(k) -5.2	(l) 21.9	(m) -5.8	(n) 22.3	(o) -6.6	(p) 22.9	(q) 22.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	23.1	22.6	21.0	17.4	14.8	11.2	11.1	12.7	16.1	18.0	19.5	21.7
	DBStd	5.35	3.21	3.08	3.24	3.71	2.97	2.87	3.29	3.62	3.56	4.20	3.60	3.46
	HDD10.0	64	0	0	0	1	3	19	26	13	1	1	1	0
	HDD18.3	989	3	4	12	61	114	213	224	177	87	57	28	9
	CDD10.0	2770	406	353	341	223	152	56	61	98	183	251	286	363
	CDD18.3	655	151	124	94	33	6	0	0	4	19	48	63	114
	CDH23.3	6340	1396	1020	784	293	69	4	7	66	256	571	711	1164
CDH26.7	2639	639	428	304	84	8	0	0	13	77	235	295	555	
Wind	WSAvg	3.7	4.2	3.8	3.6	3.1	2.9	3.2	3.2	3.3	3.7	4.2	4.5	4.7
Precipitation	PrecAvg	301	36	44	46	31	10	9	12	12	8	28	38	40
	PrecMax	425	138	135	141	78	61	44	75	55	72	106	122	148
	PrecMin	147	0	0	5	1	0	0	0	0	0	1	0	0
	PrecStd	76	32	32	30	21	14	10	17	16	14	28	29	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.6	37.0	34.8	31.5	27.6	23.4	24.5	28.7	31.9	35.2	35.7	37.9
		MCWB	17.9	17.5	16.9	15.8	13.6	10.9	10.3	11.9	13.5	15.6	16.0	16.4
	2%	DB	34.8	33.8	32.4	29.0	25.2	21.3	22.3	25.2	28.9	32.1	32.9	34.9
		MCWB	17.9	17.9	17.3	15.4	12.5	10.0	9.8	11.0	13.0	14.8	15.3	16.9
	5%	DB	32.5	31.3	30.1	26.6	23.1	19.7	20.3	22.5	26.3	29.2	30.3	32.4
		MCWB	17.7	18.0	17.0	14.7	12.2	9.3	9.3	10.3	12.5	14.2	15.0	16.6
	10%	DB	30.2	29.1	27.8	24.2	21.0	17.7	17.9	19.9	23.6	26.4	27.7	29.7
		MCWB	17.6	17.9	16.6	14.2	11.6	8.6	8.5	9.4	11.7	13.8	14.8	16.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.6	21.1	19.9	17.8	15.0	12.1	11.5	13.9	15.7	17.5	18.4	19.5
		MCDB	29.7	29.6	29.2	25.0	22.2	20.4	21.4	21.9	26.4	27.9	28.1	29.2
	2%	WB	19.8	20.0	19.0	16.9	14.1	11.0	10.6	12.7	14.5	16.5	17.2	18.7
		MCDB	27.7	27.9	27.9	24.1	21.7	19.1	19.7	20.7	24.1	26.5	27.5	28.2
	5%	WB	19.2	19.3	18.3	16.1	13.3	10.1	9.8	11.7	13.6	15.7	16.4	18.1
		MCDB	27.2	26.7	26.1	23.7	20.2	17.5	18.6	18.9	22.2	25.4	26.3	27.4
	10%	WB	18.7	18.8	17.6	15.2	12.5	9.2	9.0	10.6	12.7	14.8	15.6	17.4
		MCDB	26.6	25.7	24.7	21.6	18.9	16.1	16.2	17.8	21.3	24.1	25.0	26.1
Mean Daily Temperature Range	5% DB	MDBR	13.9	13.2	13.4	13.0	12.9	12.8	13.4	13.5	14.3	14.7	14.8	14.6
		MCDBR	17.6	16.9	16.5	16.1	15.6	15.5	16.1	17.3	18.3	19.2	19.4	19.4
		MCWBR	4.3	4.3	4.8	5.2	6.1	7.4	7.4	7.4	6.9	6.1	5.6	4.9
	5% WB	MCDBR	14.1	13.4	14.0	13.5	12.8	13.8	14.5	14.0	15.4	16.3	15.8	14.8
		MCWBR	4.3	4.3	4.7	4.7	5.5	6.8	7.1	5.9	6.2	5.8	5.6	4.8
Clear-Sky Solar Irradiance	taub	0.337	0.342	0.328	0.301	0.283	0.270	0.270	0.312	0.338	0.348	0.327	0.331	
	taud	2.518	2.506	2.519	2.575	2.600	2.605	2.594	2.452	2.367	2.382	2.493	2.525	
	Ebn at Noon	1001	977	956	932	902	893	909	904	929	959	1005	1011	
	Edh at Noon	112	110	102	86	74	69	73	95	116	123	114	112	
All-Sky Solar Radiation	RadAvg	7.40	6.72	5.69	4.40	3.49	3.08	3.35	4.18	5.42	6.48	7.34	7.85	
	RadStd	0.37	0.36	0.32	0.19	0.14	0.17	0.19	0.25	0.33	0.33	0.38	0.46	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page